

FORENSIC ANALYSIS REPORT

Image: jo-favorites-usb-2009-12-11.E01 • 2026-02-23 00:33:47 • Model: meta-llama/llama-3.1-8b-instruct

Risk Level: MEDIUM

EXECUTIVE SUMMARY

This analysis reveals potential timestomping, shadow deletion, and log clearing activities. Further investigation is required to confirm these findings and determine the extent of anti-forensic activity.

Analysis completed in 4.8 seconds. AI analysis identified 4 findings. Anti-forensic analysis detected 260 potential indicators.

AI ANALYSIS FINDINGS

Finding #1: timestomping	MEDIUM	80% conf
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Evidence: Future timestamps found: 1, Sample future dates: ['USN journal missing']

Explanation: The presence of future timestamps indicates potential timestomping activity. The USN journal missing entry suggests tampering with file system metadata.

Recommendation: Investigate the file with future timestamps to determine the original timestamp and potential modifications.

Finding #2: shadow_deletion	HIGH	90% conf
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Evidence: \$Extend metadata exists: False, System Volume Information folder: False

Explanation: The absence of \$Extend metadata and the System Volume Information folder strongly indicates that Volume Shadow Copies (VSS) are disabled or have been removed, which is a common anti-forensic technique.

Recommendation: Verify the absence of Volume Shadow Copies and investigate potential deletion tools.

Finding #3: deletion	LOW	40% conf
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Evidence: Orphaned files: 1

Explanation: The presence of an orphaned file suggests that a file may have been deleted, but the absence of other deleted files indicates a targeted or limited deletion.

Recommendation: Investigate the orphaned file to determine its original location and potential deletion method.

Finding #4: log_clearing	MEDIUM	70% conf
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Evidence: Windows Event Logs (.evtx): 0

Explanation: The absence of Windows Event Logs suggests that logs may have been cleared, potentially to conceal malicious activities.

Recommendation: Investigate the absence of logs and potential log clearing tools.

ANTI-FORENSIC ANALYSIS

TIMESTAMPING DETECTION [256 indicators]

Detects manipulation of file timestamps

- Partition 0: Zero timestamp detected: r/r 3: JO FAVORITE (Volume Label Entry) 2009-11-20 09:30:34 (IST) 0000-00-00 00:
 - Partition 0: Zero timestamp detected: r/r 5: ._Trashes 2009-11-17 16:45:26 (IST) 2009-11-20 00:00:00 (IST) 0000-00-00
 - Partition 0: Zero timestamp detected: r/r 6: DSC00009.JPG 2009-11-11 09:16:56 (IST) 2009-11-24 00:00:00 (IST) 0000-00-
 - Partition 0: Zero timestamp detected: d/d 8: .Trashes 2009-11-17 16:45:26 (IST) 2009-11-17 00:00:00 (IST) 0000-00-00 0
 - Partition 0: Zero timestamp detected: + d/d * 133: ._02 2009-11-20 09:25:26 (IST) 2009-11-20 00:00:00 (IST) 0000-00-00
 - Partition 0: Zero timestamp detected: + r/r * 135: ._502 2009-11-20 09:25:28 (IST) 2009-11-20 00:00:00 (IST) 0000-00-0
 - Partition 0: Zero timestamp detected: + r/r * 136: ._502~1 2009-11-17 16:45:28 (IST) 2009-11-17 00:00:00 (IST) 0000-00-
 - Partition 0: Zero timestamp detected: + r/r * 138: ._501 2009-11-17 16:45:28 (IST) 2009-11-17 00:00:00 (IST) 0000-00-0
 - Partition 0: Zero timestamp detected: d/d * 10: .fseventsd 2009-11-17 16:45:26 (IST) 2009-11-17 00:00:00 (IST) 0000-00
 - Partition 0: Zero timestamp detected: d/d 13: .Spotlight-V100 2009-11-17 16:45:26 (IST) 2009-11-17 00:00:00 (IST) 0000
 - Partition 0: Zero timestamp detected: + d/d 646: Store-V1 2009-11-17 16:45:26 (IST) 2009-11-17 00:00:00 (IST) 0000-00-
 - Partition 0: Zero timestamp detected: ++ r/r 775: VolumeConfig.plist 2009-11-17 16:45:28 (IST) 2009-11-20 00:00:00 (IST) (IS
 - Partition 0: Zero timestamp detected: ++ d/d 777: Stores 2009-11-17 16:45:28 (IST) 2009-11-17 00:00:00 (IST) 0000-00-0
 - Partition 0: Zero timestamp detected: +++ d/d 1544: BC650BA8-C5B5-4A64-8BDB-18A9533BF1D7 2009-11-17 16:45:28 (IST) 200
 - Partition 0: Zero timestamp detected: ++++ r/r 1669: psid.db 2009-11-20 09:30:34 (IST) 2009-11-20 00:00:00 (IST) 0000-
- ... and 241 more

SHADOW COPY DELETION [Clear]

Evidence of VSS deletion attempts

No indicators found in this category.

HIDDEN DATA STREAMS (ADS) [Clear]

Alternate Data Streams detection

No indicators found in this category.

LOG CLEARING EVIDENCE [2 indicators]

Evidence of log file clearing

- Partition 0: Few or no log files found - possible clearing
- Partition 0: No Windows event log files (.evtx) found

REGISTRY TAMPERING [Clear]

Registry modification indicators

No indicators found in this category.

FILE DELETION TRACKING [1 indicator]

Deleted file recovery opportunities

- Partition 0: Orphaned files found (deleted but recoverable): 1 entries

MFT ANOMALIES [1 indicator]

Master File Table irregularities

- Partition 0: MFT contains 4 system file entries

SUMMARY STATISTICS

Category	Indicators
Timestamping	256
Shadow Copy Deletion	0
Hidden Streams (ADS)	0
Log Clearing	2
Registry Tampering	0
File Deletion	1
MFT Anomalies	1
TOTAL	260

Overall Anti-Forensic Risk Level: HIGH

RECOMMENDATIONS

1. Investigate file with future timestamps, verify Volume Shadow Copies deletion, and examine log clearing tools and methods.

Additional Anti-Forensic Investigation Steps:

- Conduct detailed timeline analysis around detected anomalies.
- Examine Windows Event Logs for evidence of vssadmin or wevtutil commands.
- Analyze \$MFT and \$UsnJrnl for file system manipulation evidence.
- Review Registry hives for persistence mechanisms and tool artifacts.
- Search for known anti-forensic tool signatures and artifacts.